PROJECT 3 · PART A

Exploring the Power of Flow Models

From noisy pixels to guided edits: sampling, reconstruction, and optical illusions with a pretrained image model.

This project explores how a flow model moves an image from noise toward a coherent result. I begin with a photograph and compare three ways to denoise it, then use text guidance to generate portraits, edit familiar shapes, and combine two interpretations in one image.

Model: PixNerd-XXL-P16-T2I · 512 × 512 image space · pretrained inference.

Conventions: RGB inputs in [–1, 1]; time runs from noise at 0 to image at 1. Euler step size h = 0.02. Guided experiments use w = 7 and an empty unconditional prompt.

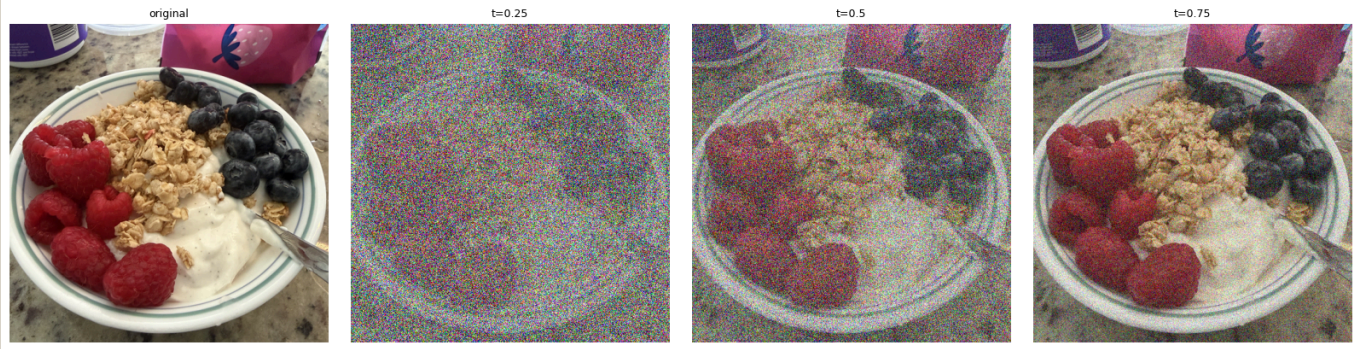
PART A · TRAINING AND SAMPLING

1.1 Constructing noisy training inputs

I use a photo of a yogurt bowl with raspberries, blueberries, and granola throughout the denoising experiments. It contains smooth regions, strong boundaries, and fine texture, so it makes the difference between removing noise and preserving detail easy to see.

$$x_t = (1 - t)\varepsilon + tx_1, \quad \varepsilon \sim \mathcal{N}(0, I)$$

The interpolation scales both the noise and the clean image. At t = 0 the input is entirely noise; at t = 1 it is the original image. Holding the image and noise realization fixed across times makes the comparison about the noise level. Increasing t makes the bowl easier to recognize, but the granola and berry surfaces are still noisy at t = 0.75.



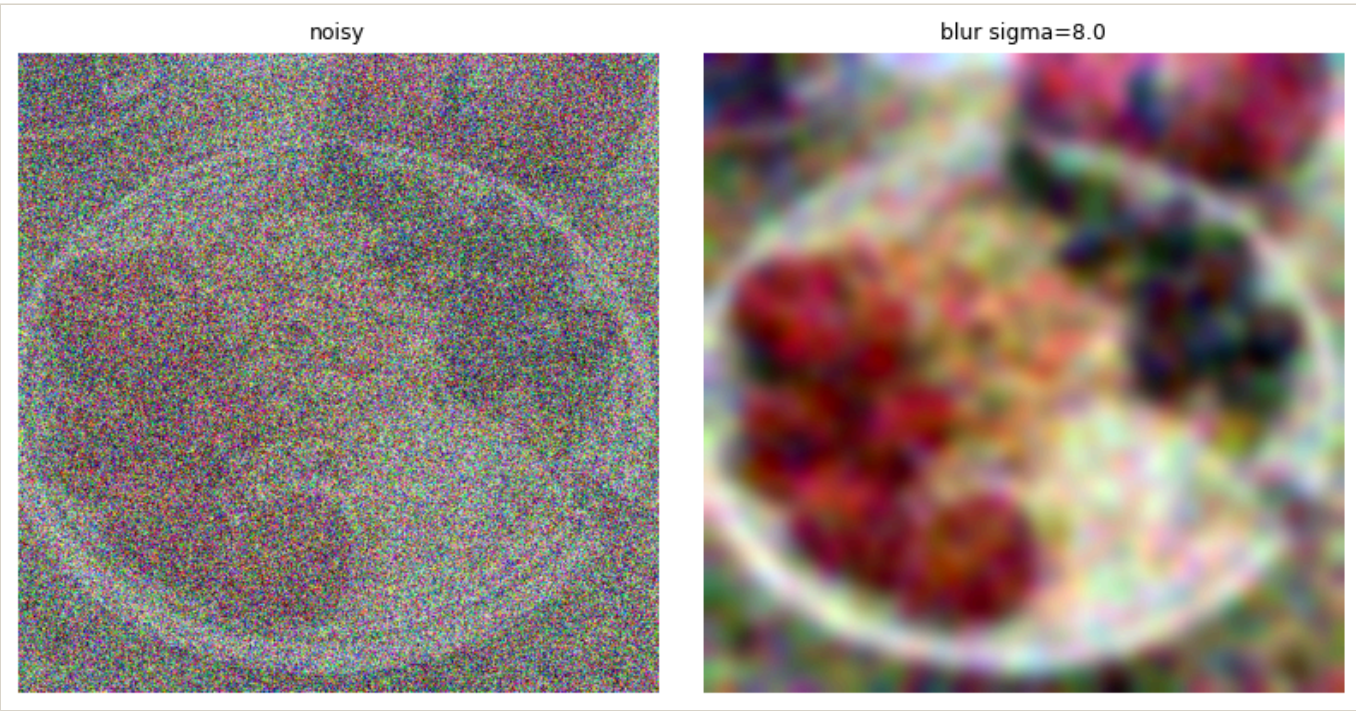
Original · t = 0.25 · t = 0.50 · t = 0.75

1.2 Classical denoising

Gaussian blur provides a baseline that does not use the learned model. Because interpolation attenuates the image by t, the baseline first divides by t and then applies the blur. This restores the signal amplitude but also amplifies the remaining noise.

$$\hat{x}_1 = G_\sigma(x_t / t)$$

The displayed blur strengths decrease with the noise level: σ = 8, 4, and 2 for t = 0.25, 0.50, and 0.75. Stronger averaging suppresses more noise, at the cost of texture. At t = 0.25 the fruit becomes broad patches of color; at t = 0.75 a smaller blur keeps the berry outlines and more of the granola.



t = 0.25 · Noisy input and Gaussian blur, σ = 8

t = 0.50 · Noisy input and Gaussian blur, σ = 4

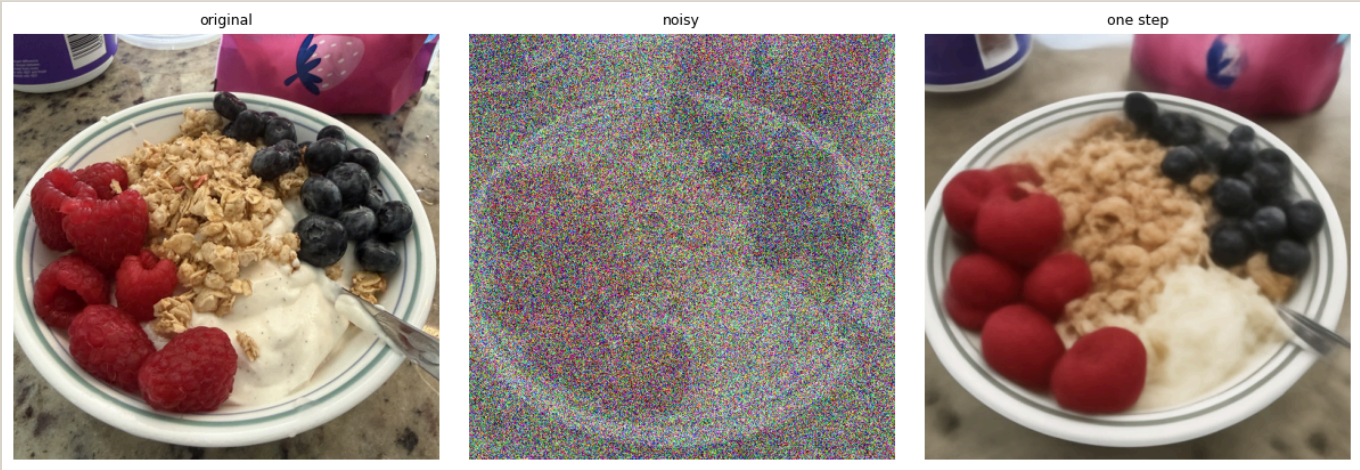
t = 0.75 · Noisy input and Gaussian blur, σ = 2

1.3 One-step denoising

The flow model predicts a velocity: a direction of change in image space. A single prediction can be extrapolated over the remaining interval to estimate the clean endpoint. The text condition for this experiment is “a high quality photo”.

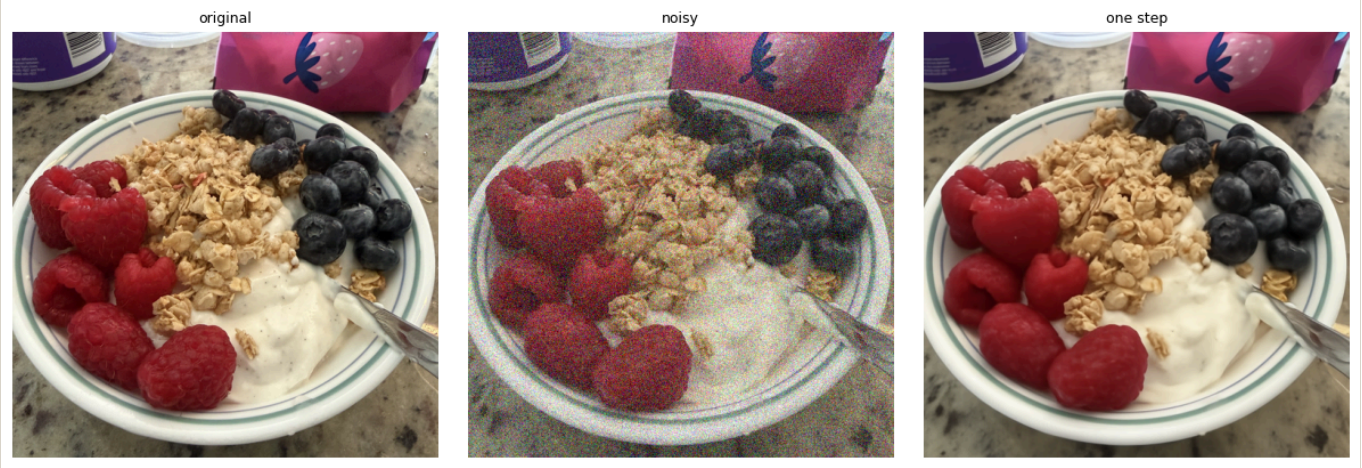
$$\hat{x}_1 = x_t + (1 - t)v_\theta(x_t, t, c)$$

This produces much more recognizable food than Gaussian blur at the same noise level. However, the low-t estimates simplify the raspberries into unusually smooth red shapes. The model can infer a plausible object without recovering its exact texture. At t = 0.75, more image evidence survives, and the estimate stays much closer to the original.



t = 0.25 · Original, noisy input, and one-step estimate

t = 0.50 · Original, noisy input, and one-step estimate



t = 0.75 · Original, noisy input, and one-step estimate

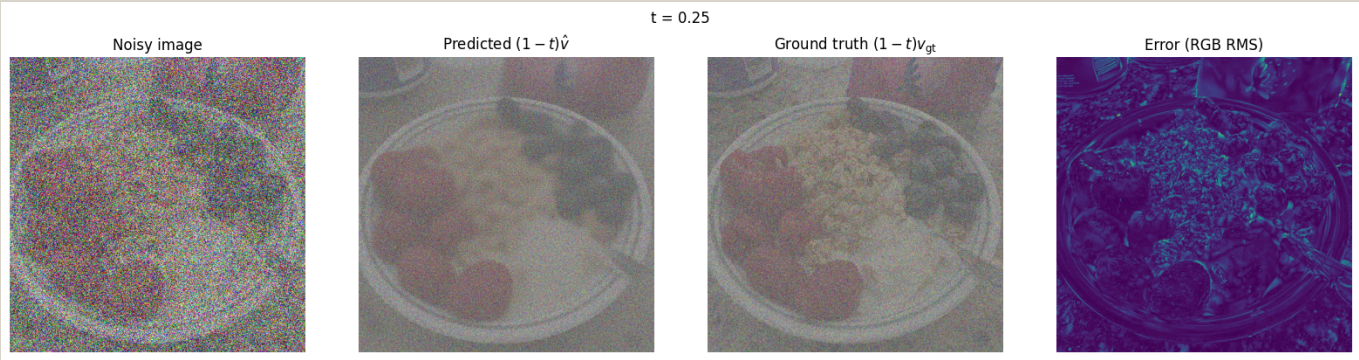
1.4 Visualizing velocity

Looking at the update itself helps explain the one-step result. For the known image/noise pair, the exact velocity is $x_1 - \varepsilon$. Multiplying by the remaining time gives the change needed to reach the original image. The comparison below is between these endpoint updates, rather than raw velocities.

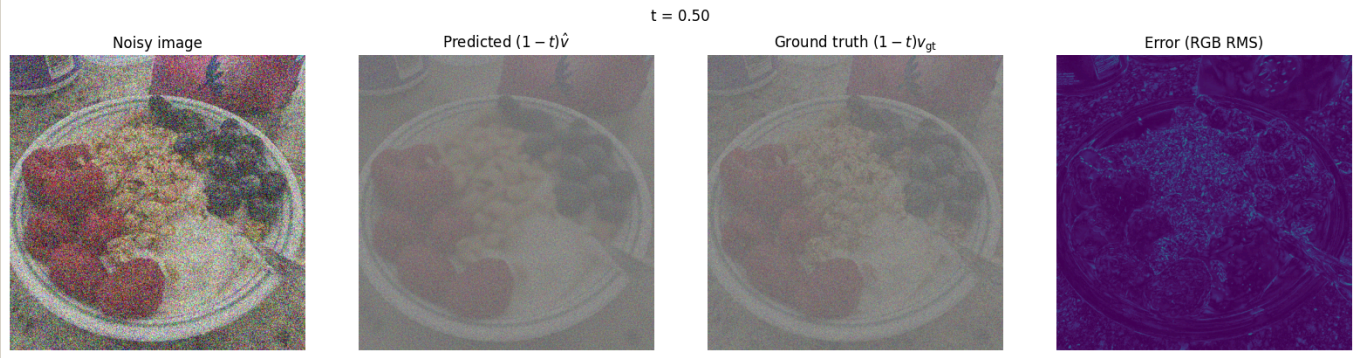
$$\hat{\Delta} = (1 - t)v_\theta \quad \cdot \quad \Delta_{gt} = x_1 - x_t$$
$$E = \sqrt{\text{mean}_{RGB}[(\hat{\Delta} - \Delta_{gt})^2]}$$

Original image · Reference for all three times

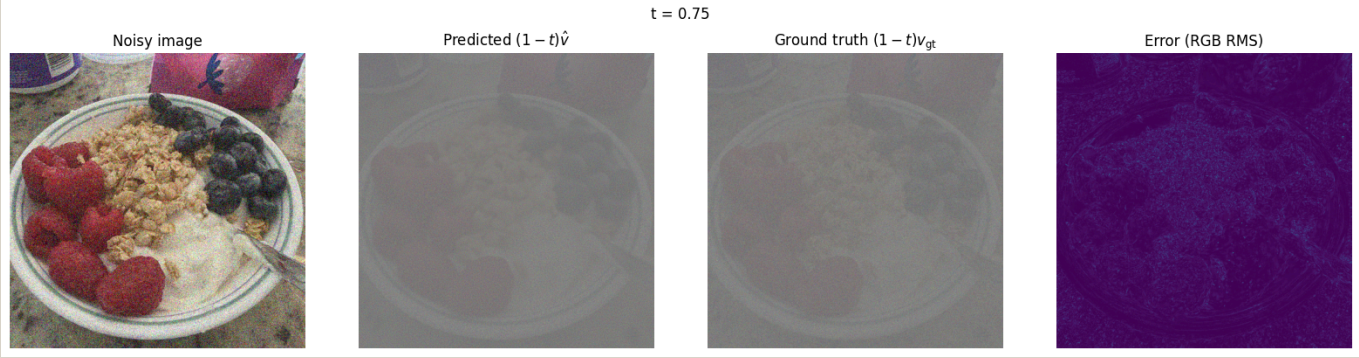
Zero update is represented by middle gray. A common symmetric scale for the updates and common linear limits for the error maps allow comparisons across times. The RMS error is computed across the RGB channels before display scaling or clipping, so positive and negative channel errors cannot cancel.



t = 0.25 · Noisy input, predicted update, ground-truth update, and RGB RMS error



t = 0.50 · Noisy input, predicted update, ground-truth update, and RGB RMS error



t = 0.75 · Noisy input, predicted update, ground-truth update, and RGB RMS error

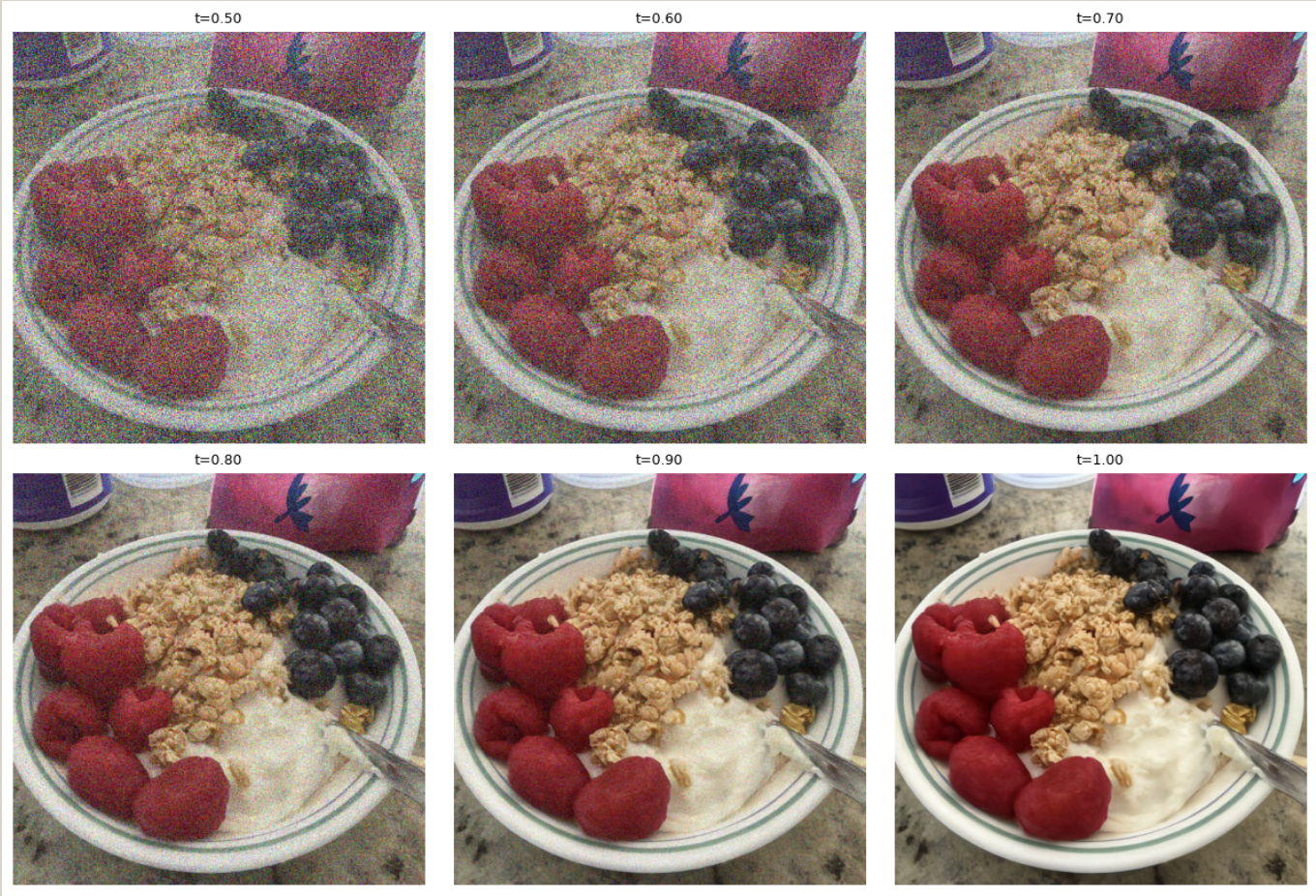
The brightest errors trace detailed regions and boundaries, including the granola and the rim of the bowl. The t = 0.75 map is noticeably darker. This reflects both a more informative input and a shorter remaining update; it should not be read as a direct comparison of unscaled velocity error.

1.5 Iterative denoising with Euler sampling

A single extrapolation assumes that the initial velocity remains useful all the way to the endpoint. Euler sampling instead takes a short step and evaluates the model again at the updated image and time. The direction can then change as the image becomes clearer.

$$x_{t+h} = x_t + h\, v_\theta(x_t, t, c)$$

Starting at t = 0.50 with h = 0.02 gives 25 updates. The trajectory shows the initial state and every fifth update, including the final result. In a general sampler, the last step is shortened to min(h, 1 – t), so integration ends at exactly t = 1.



Euler trajectory · t = 0.50, 0.60, 0.70, 0.80, 0.90, 1.00

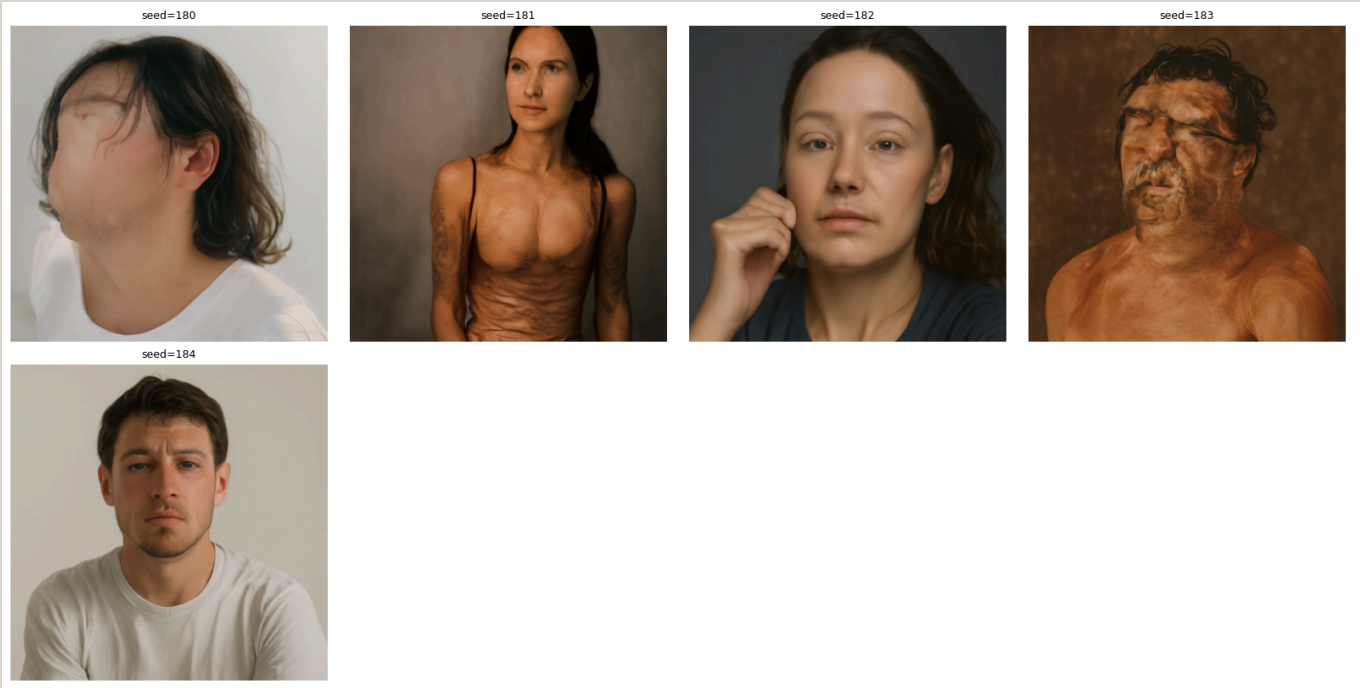
Comparing the denoisers

Same t = 0.50 input · Original, Gaussian blur, one-step estimate, and Euler result

Gaussian blur leaves the scene soft. One-step denoising recovers the arrangement but smooths the raspberry texture. Euler restores more of the small surface detail and the granola's structure. Its cost is 25 model evaluations instead of one; the sharper appearance also does not guarantee pixel-perfect recovery.

1.6 Generating images from noise

The same sampler can start from pure Gaussian noise at t = 0 and integrate to t = 1. There is now no input photograph to reconstruct. The prompt “a high quality photo” supplies only a broad preference, leaving the model to choose the subject.



Five samples · Seeds 180–184 · Prompt: “a high quality photo”

All five samples are portraits, even though the prompt does not request people. Their quality varies substantially: seed 180 has a largely missing face, while seed 183 has an irregular, painterly surface. Seeds 182 and 184 look more like ordinary photographs. This small sample suggests a strong portrait preference for this generic prompt; it is not a measurement of the model’s overall subject distribution.

PART A · CONDITIONAL GENERATION AND GUIDANCE

2.1 Classifier-free guidance

Classifier-free guidance compares the conditional prediction with a prediction using an empty prompt. Extrapolating their difference strengthens the direction associated with the text condition. Here the guidance convention is conditional velocity plus w times the difference, so w = 0 already means ordinary conditional sampling.

$$v_{CFG} = v_c + w(v_c - v_u), \quad w = 7$$

The results below use the same seed labels, 180–184, as the preceding samples, with the same generic photo prompt. Comparing matched seeds helps isolate the effect of guidance instead of comparing unrelated random draws.

Five guided samples · Seeds 180–184 · w = 7

The guided outputs have much more coherent faces and consistent portrait lighting. They also converge on a similar polished headshot style. Guidance improves the visible quality here, but does not broaden the subject matter: every result is still a portrait. The change can be large enough to alter the apparent identity and composition for a given seed.

PART A · CREATIVE APPLICATIONS

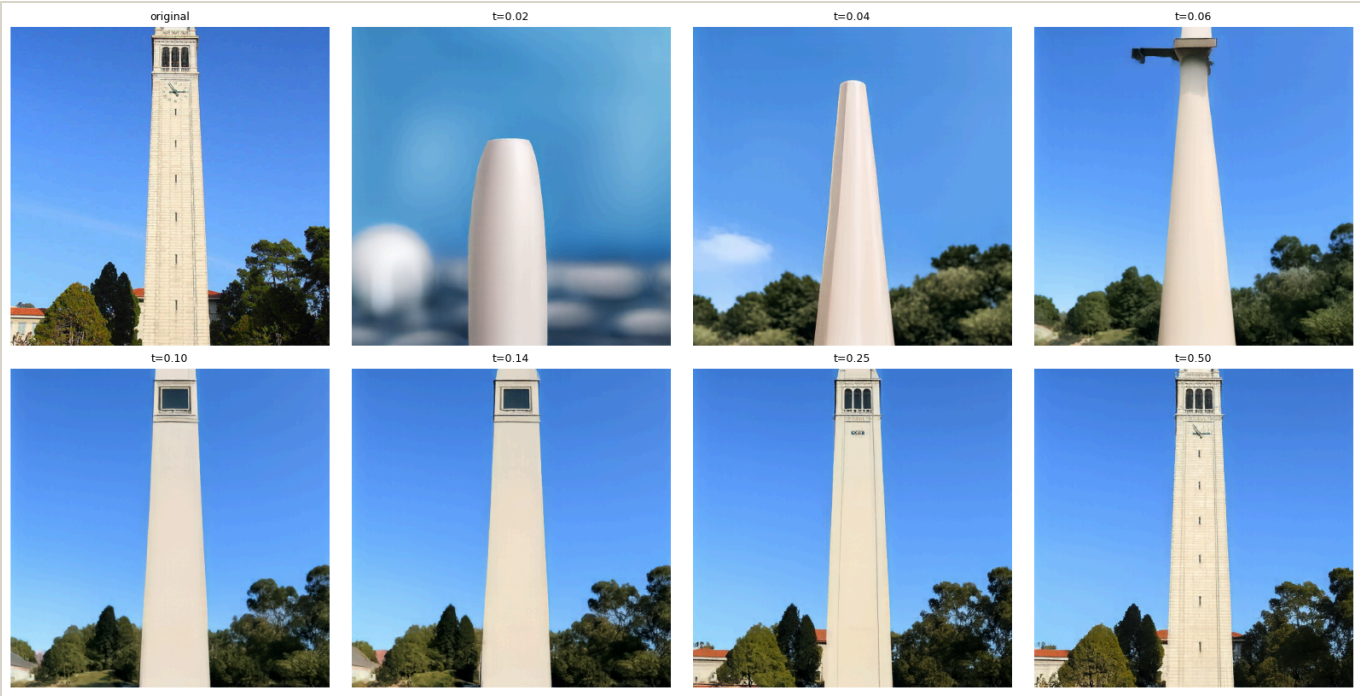
3.1 Image-to-image editing

Image-to-image editing begins partway along the noise-to-image path: mix an input with noise at a chosen starting time, then integrate to t = 1. Small starting times leave less input information and permit larger changes. Large starting times preserve more of the source. The editing sweeps use h = 0.02, guidance w = 7, and an empty unconditional prompt.

The generic prompt is “a high quality photo”. These three inputs show how much the retained geometry can constrain the result without describing a specific edit.

Each grid reads left to right, top row then bottom row: original, t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50.

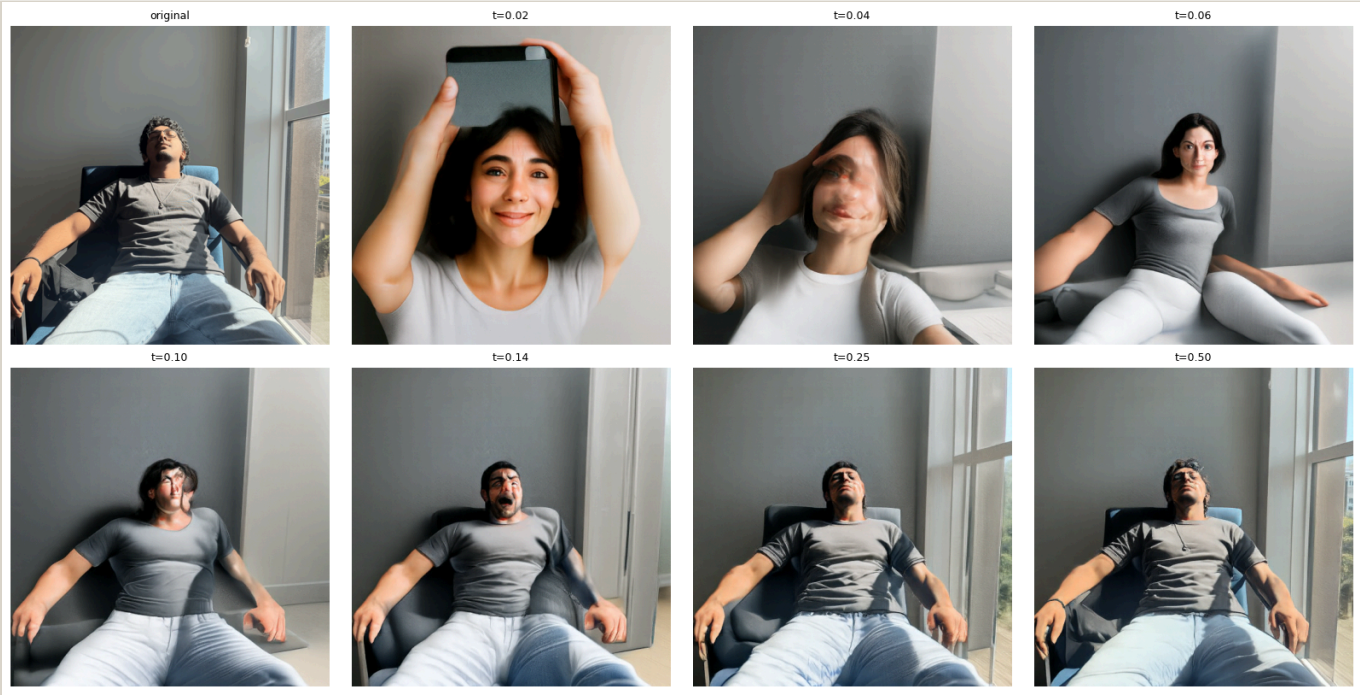
Campanile



Campanile · Original and seven starting times

At t = 0.02 the tower becomes a rounded vertical object. The long silhouette and blue sky survive before the architectural details do. By t = 0.25 the belfry is recognizable, and t = 0.50 closely preserves the original tower.

Sunlit portrait



Sunlit portrait · Original and seven starting times

The lowest starting times produce different people and poses. At t = 0.10 and above, the reclining pose, gray shirt, and room layout become increasingly stable. Preserving composition is easier than preserving the exact person when little of the input remains.

Pizza

Pizza · Original and seven starting times

The pizza disappears entirely at the first few times, giving way to people. At t = 0.10, human forms and the circular pizza overlap awkwardly. By t = 0.14 it is recognizable as pizza again. This is a useful failure case: a generic prompt cannot reliably preserve the subject when the input is mostly noise.

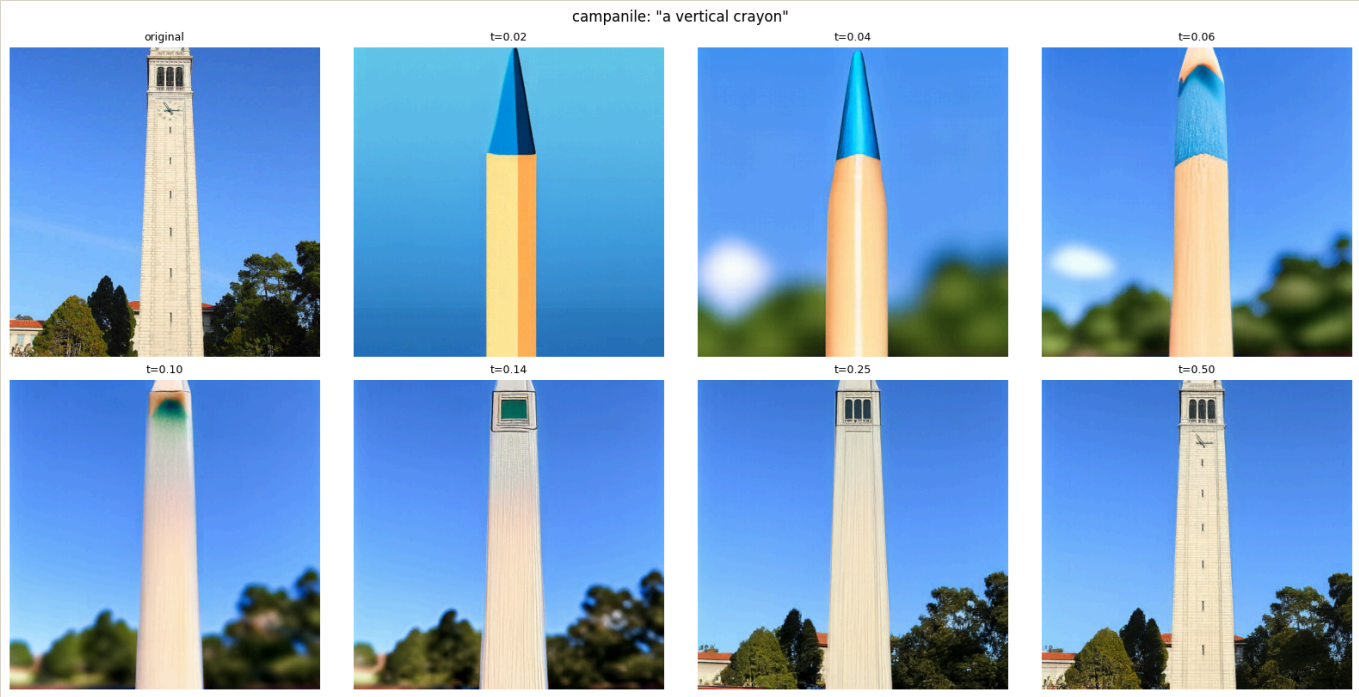
3.2 Text-guided image editing

Replacing the generic prompt with a desired edit adds a second constraint: the result should follow the text while retaining some of the source image. I pair each input with an edit related to its geometry—a tower and crayon, a seated person and jacket, and a pizza and clock.

Each grid reads left to right, top row then bottom row: original, t = 0.02, 0.04, 0.06, 0.10, 0.14, 0.25, 0.50.

Campanile → crayon

> **Prompt** “a vertical crayon”



Vertical crayon · Original and seven starting times

The narrow tower provides a strong shape cue. The first three edits produce a pointed drawing tool with yellow and blue coloring; by t = 0.10 the tower begins to reappear. The prompt transfers the silhouette successfully, although the early results look closer to a pencil than a wax crayon.

Sunlit portrait → down jacket

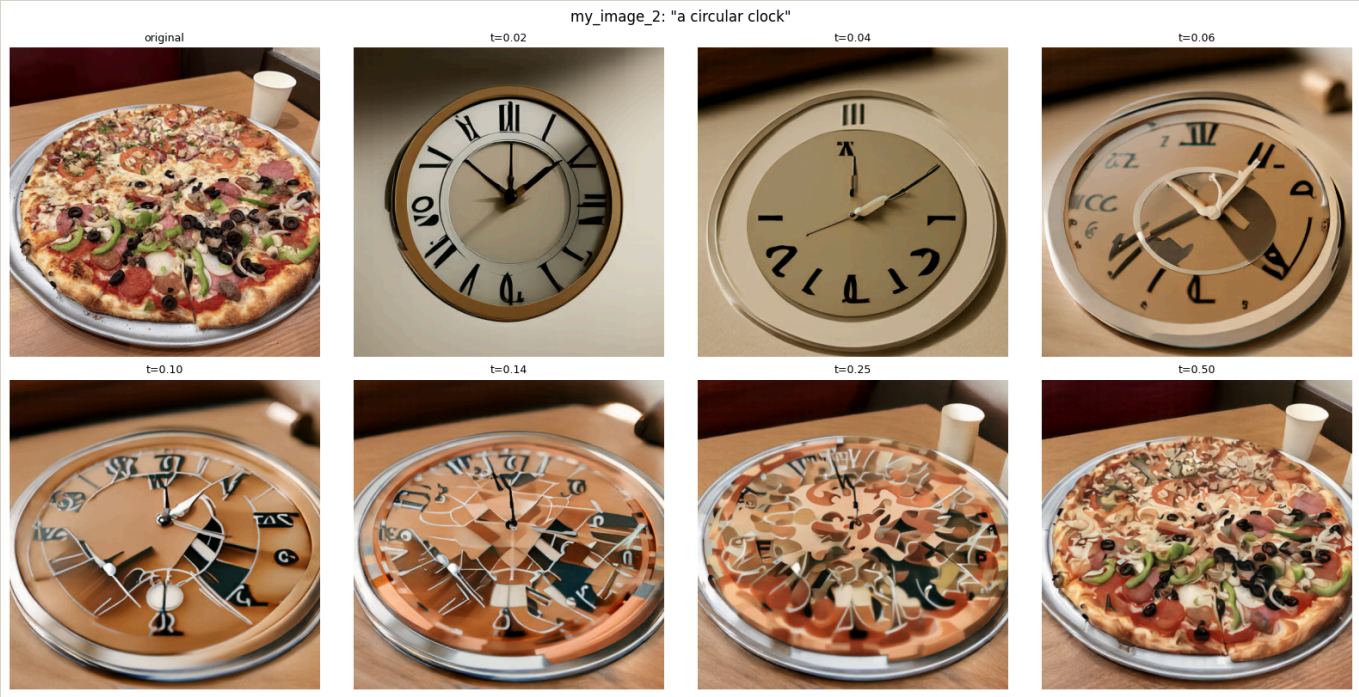
> **Prompt** “a person wearing a thick down jacket”

Down jacket · Original and seven starting times

At t = 0.02–0.06, the jacket dominates and the person changes substantially. Around t = 0.10–0.14, the padded jacket and seated composition coexist. By t = 0.50 the original shirt largely returns. The middle of the sweep gives the most useful balance between adding clothing and retaining the scene.

Pizza → clock

> **Prompt** “a circular clock”



Circular clock · Original and seven starting times

The circular outline makes this a natural shape match. Small starting times give recognizable clock faces; intermediate times preserve the tabletop view while blending clock numerals with toppings. At t = 0.25, the pizza is still visible underneath the clock features, and t = 0.50 mostly restores the food.

3.3 Visual anagrams

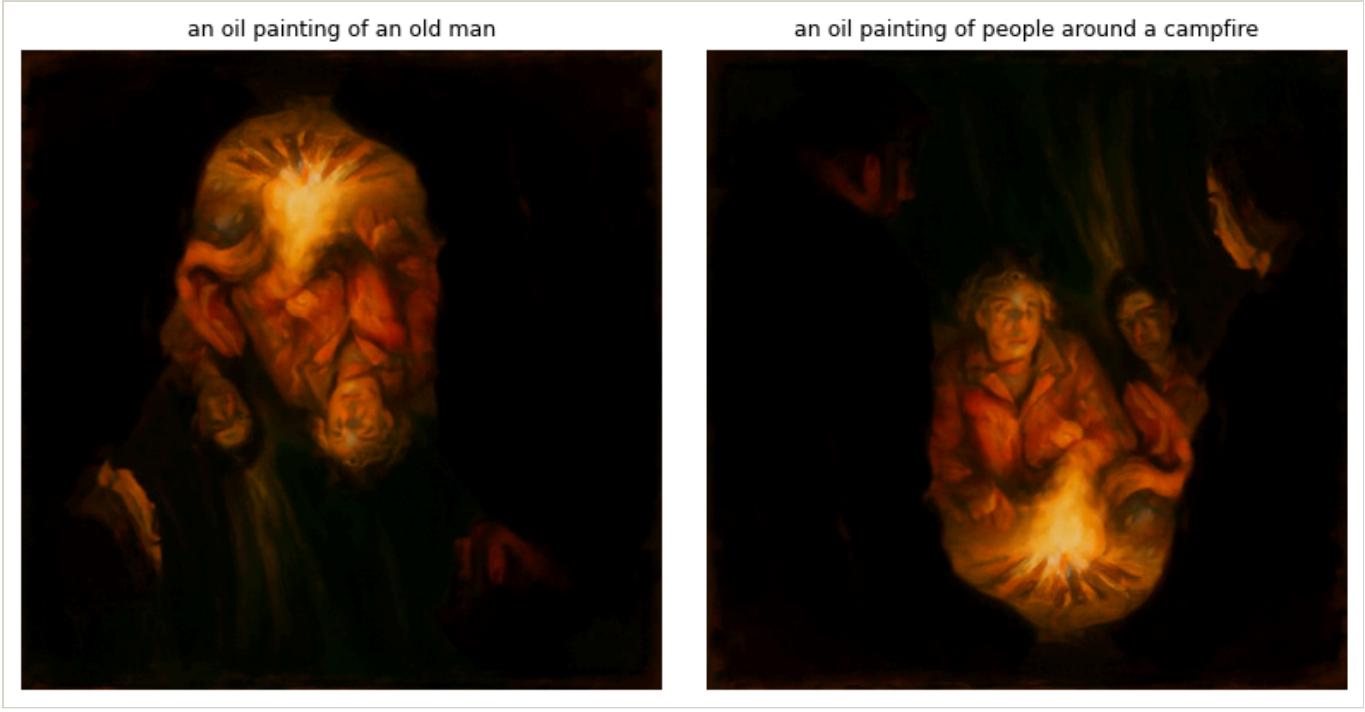
For an upside-down illusion, one prompt is evaluated on the current image and another on its 180° rotation. The second predicted velocity must be rotated back before averaging, so both updates refer to the same pixel coordinates. Averaging the guided velocities at every Euler step encourages a single image to support both readings.

$$v = \tfrac{1}{2}[\mathrm{CFG}(x, t, p_1) + R(\mathrm{CFG}(R(x), t, p_2))]$$

R rotates both spatial axes by 180°.

Old man ↔ campfire

> **Upright** “an oil painting of an old man”
> **Rotated** “an oil painting of people around a campfire”

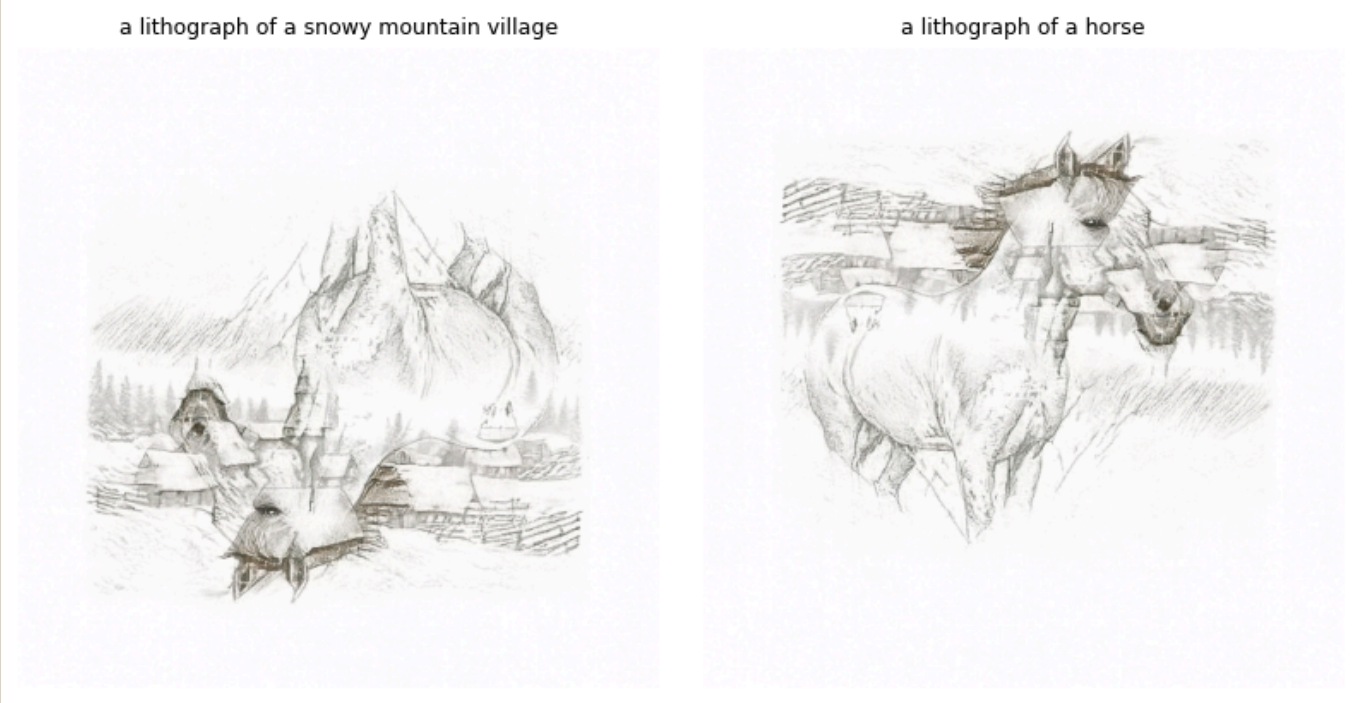


The same image · Upright and rotated 180°

The warm highlights serve two purposes: facial structure upright, and firelight when rotated. The face remains a strong competing interpretation in the flipped view, so the campfire reading is more ambiguous. The shared dark palette makes the two prompts visually compatible.

Snowy village ⇔ horse

> **Upright** “a lithograph of a snowy mountain village”
> **Rotated** “a lithograph of a horse”



The same image · Snowy village and rotated horse

The pale background and sparse linework let mountain slopes and rooftops double as the horse’s contours. The horse is especially legible in the rotated view, while the village view depends more on interpreting the small buildings and landscape marks.

3.4 Hybrid images

A hybrid uses two prompts at the same image and time, but combines different frequency bands of their guided velocity predictions. The low-pass component sets broad structure; the high-pass residual contributes fine detail. Their sum becomes the velocity for the next Euler step.

$$v = G_\sigma(v_{low}) + [v_{high} - G_\sigma(v_{high})]$$

The high-pass term is the prediction minus its Gaussian blur. This isolates detail rather than adding a second full image. A broad Gaussian filter separates the two scales; the assignment suggests a 265-pixel kernel with $\sigma = 16$ at 512 × 512. Each result is shown at two display sizes to make the change in interpretation easier to inspect.

Waterfall / skull

High-frequency subject: Waterfall · Low-frequency subject: Skull

Near view · Waterfall

Distant view · Skull

The waterfall’s vertical streams and surrounding trees are visible at full size. When reduced, the broad dark regions begin to read as facial cavities within a pale skull-like outline. The skull interpretation is subtle, so the near/far comparison is more informative than the full-size image alone.

Forest / cat

High-frequency subject: Forest · Low-frequency subject: Cat

Near view · Forest

Distant view · Cat

Up close, the thin trunks and branches read as a forest. The broader silhouette above the trees suggests an animal head and ears when viewed small, although the cat is less immediately recognizable than the forest. This illustrates the difficulty of making both frequency bands equally convincing.

Reflection

The denoising experiments show a progression from local averaging to learned reconstruction: blur removes detail along with noise, while the flow model can recover recognizable structure and add plausible texture. Repeated Euler updates improve the yogurt bowl's fine detail over a single extrapolation.

The editing experiments make the tradeoff more visible. Less input information gives the prompt more freedom, but also makes identity and subject preservation unreliable. Guidance sharpens the generic samples into convincing portraits without escaping the portrait preference. The anagrams and hybrids push this further: one result has to satisfy competing interpretations, and the weaker interpretation can remain ambiguous.

Assignment: CS 180 · Project 3, Part A.